

Nafiz Khan

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Profile

I am a B.Sc. Graduate in Computer Science and Engineering from BRAC University with a focus on software engineering and applied machine learning. I am a full-stack developer specializing in the MERN stack, with experience in building scalable, production-ready web applications. I completed undergraduate research on anchor-guided weight repair for stabilizing pretrained language models under quantization and weight noise. With a strong foundation in data structures, algorithms, and system design, I aim to develop robust, efficient systems that bridge modern software engineering and trustworthy AI.

Skills

- **Languages:** Python, C, JavaScript, SQL, Solidity, Assembly, TypeScript
- **Frameworks & Libraries:** MERN Stack, Socket.IO, PyTorch, Scikit-learn, Web3.js, Prisma ORM, Langchain
- **Tools & Platforms:** Docker, Git, GitHub, Supabase, Hyperledger Fabric, OAuth2, Passport.js, n8n, Puppeteer
- **AI & Processing:** NLP, Groq AI, Compromise.js, pdf-parse, mammoth, Regex
- **Other:** Linux, Figma, Arduino, STM32, Raspberry Pi, Web Crypto API

Education

- BRAC University** — BSc in Computer Science and Engineering Jan 2022 – Jan 2026
- GPA: 3.65 / 4.00
- Adamjee Cantonment College** — Higher Secondary Certificate (HSC) 2018 – 2020
- GPA: 5.00 / 5.00
- Bangladesh International School and College** — Secondary School Certificate (SSC) 2004 – 2018
- GPA: 5.00 / 5.00

Experience

- AI & Automation Specialist**, Softvence Agency — Dhaka, Bangladesh Apr 2026 – June 2026
- Design and deploy automation workflows using n8n based on client requirements.
 - Build API integrations and AI-driven automations with testing and optimization.
 - Delivered 7+ high-performance production applications for clients across diverse domains.
- Executive, Network & IT**, Wintel Limited — Dhaka, Bangladesh Dec 2025 – Feb 2026
- Supported network infrastructure maintenance and IT operations within the Technical Team.
 - Assisted in troubleshooting, system monitoring, and technical documentation.
- IT Instructor**, Zentorra — Dhaka, Bangladesh Feb 2025 – Jul 2025
- Delivered IT training sessions covering fundamental and practical concepts.

Research Experience

- Undergraduate Researcher**, BRAC University, CSE Jan 2025 – Jan 2026
- Title:** Anchor-Guided Repair: Enhancing Stability of Compromised Pretrained Language Models Against Low-Precision and Weight Noise Attacks [[Google Scholar](#)]
- Proposed **Anchor-Guided Repair**, a post-deployment defense that restores compromised pretrained language models using anchor-based regularization without requiring access to proprietary training data.
 - Designed a joint optimization framework combining language modeling loss with anchor loss to improve robustness against Gaussian weight noise and low-bit quantization attacks while preserving learned knowledge.
 - Evaluated the approach across multiple LLM architectures and attack scenarios, consistently recovering model stability and performance close to clean baselines.

Projects

TrackUp - Job Application Management Platform

- Built a platform for managing job applications with tracking, reminders, analytics, AI chatbot, resume parsing, and job recommendations via NLP.
- Tools Used: React.js, Tailwind CSS, Node.js, Express.js, MongoDB, Socket.IO, Docker, Vercel, Render

NavWatch - AIS Vessel Map

- Built a real-time AIS vessel tracking application that decodes live NMEA data from a TCP feed, stores vessel positions in MongoDB, and streams live updates via WebSockets to an interactive map with searchable vessel information.
- Tools Used: React.js, Node.js, Express.js, MongoDB, Mongoose, WebSocket, Leaflet, Vite

Credenza - Encrypted Credential Manager

- Client-side, zero-knowledge credential vault utilizing Web Crypto API (PBKDF2 + AES-GCM) for secure local encryption, with an integrated encrypted reconnaissance tracker for managing cybersecurity data and insights.
- Tools Used: Next.js, React, Node.js, Prisma, PostgreSQL, Web Crypto API, Vercel

ThesisFlow - Thesis Management System

- Developed a web-based platform to streamline thesis submission, progress tracking, and supervision for students and faculty.
- Tools Used: MongoDB, Express.js, React.js, Node.js (MERN stack), HTML, CSS

Multi-Agent AI Debate System with Langchain

- Developed a multi-agent AI debate platform where two specialized AI agents engage in structured arguments and rebuttals while a neutral AI judge evaluates responses based on logic, clarity, and persuasiveness, generating structured JSON-based verdicts.
- Tools Used: Python, LangChain, Groq API, Mixtral-8x7B, Google Colab, Prompt Engineering

Multimodal RAG Agent for Clothing Store

- Built a multimodal Retrieval-Augmented Generation (RAG) system that matches customer-uploaded clothing images with a product catalog using CLIP embeddings and vector similarity search, delivering product recommendations and metadata through JSON APIs.
- Tools Used: Python, FastAPI, LangChain, CLIP, ChromaDB, SQLite, Hugging Face, PyTorch, ngrok

Due Diligence Questionnaire Agent

- Built a retrieval-augmented generation (RAG) system that automates due diligence questionnaires using document indexing, semantic retrieval, and LLM-based answer generation with source citations.
- Tools: Node.js, Express, MongoDB, Redis, ChromaDB, Groq (LLaMA 3.1), HuggingFace, React, TypeScript, Tailwind CSS

Gesture Detection ML Pipeline

- Hand gesture recognition pipeline using LeapGestRecog and MediaPipe (63 features), with real-time webcam classification via a trained scikit-learn model.
- Tools Used: Python, MediaPipe, scikit-learn, OpenCV, NumPy, Pandas

Sweet Shop - E-Commerce Platform

- Built a full-stack e-commerce platform for a sweet shop with shopping cart, secure payment processing, order tracking, fraud prevention, and admin panel for product management.
- Tools Used: React.js, TypeScript, Vite, Tailwind CSS, Supabase (PostgreSQL), shadcn-ui

DisasterGuard DApp - Blockchain Disaster Recovery Training Platform

- Built a decentralized application (DApp) for training and awareness in disaster recovery, leveraging blockchain for secure certification and training records.
- Tools Used: Solidity, Ethereum, Web3.js, React.js, Ganache

Achievements

- **Academic High Distinction** | **DEAN'S LIST** (SUMMER 24 & SPRING 25), BRAC UNIVERSITY.
- **2nd Runner-up** | BRAC CSE471 Project Showcase (SUMMER 2025) for **ThesisFlow**
- Completed an **Intermediate ChatGPT** course on DataCamp, focusing on advanced prompt engineering and understanding GPT architecture.
- Completed the **Learn Blockchain and Cryptocurrency from Beginning** course on Udemy, gaining foundational knowledge in blockchain, Bitcoin, crypto exchanges, and DeFi.
- Completed the **Intro to Machine Learning** course on Kaggle, learning practical data science and ML.
- Completed the **AI Foundation** course on AI Certs, covering core AI concepts, machine learning, ethics, and business applications.
- Completed the **Intermediate SQL** course on DataCamp, mastering aggregate functions, sorting, grouping, and data presentation.